Zach Robertson

Senior Hardware Software Integration Engineer

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> SUMMARY

Senior hardware-software integration engineer for electric RV platforms. I build the verification plans, test infrastructure, internal tools and failure forensics that let engineering, manufacturing and field teams find and fix problems fast across embedded Linux, VCU firmware, CAN networks and fleet telemetry.

> EXPERIENCE

Lightship

Denver, CO · Jun 2024 – Present

Senior Hardware Software Integration Engineer

2026 – Present

- Architected and built ShipLine, Lightship's end-of-line manufacturing test platform. Worked with manufacturing to map their pain points and developed high-value tests for their stations, on FastAPI services, a PostgreSQL spine-and-detail results schema, live WebSocket station status and per-station YAML test configuration.
- Redesigned the VCU Linux image for the September 2026 release: a baked image (V16) replacing runtime commissioning, a RAUC A/B OTA pipeline and a Docker/CI builder producing drop-in UBI images, verified by a 45-test, nine-group DVP.
- Root-caused fleet WiFi/BT outages on the SD8987 module to three driver and firmware defects, exposed Bluetooth-fed TPMS data as a single point of failure for propulsion control, built a fleet-wide MCAP outage scanner and managed the supplier escalation.
- Defined hardware-in-the-loop test strategy: specified a Raspberry Pi HIL rig for VCU behavioral testing (SocketCAN, BLE/TPMS spoofing, WiFi AP control, power fault injection) and the test plan and HIL scope for the Awning V1 program.

Hardware Software Integration Engineer

Jun 2024 – 2026

- Architected the cloud CAN telemetry pipeline: S3-triggered Lambda processing, TRC conversion, MCAP validation, DynamoDB version tracking and chat alerting for fleet-wide log triage.
- Founding member of the vehicle triage team across 1,000+ vehicle issues: designed its workflow and playbooks, wrote technician troubleshooting guides, and built internal tools (CAN utilities, MF4-to-TRC converters, ECU flashing wrappers) adopted by engineering and production.
- Led a 21-day telematics supplier transition to Owasys embedded Linux, and wrote bring-up and validation frameworks for HV, LV, HVAC, tub and canopy subsystems, including the Bender ISO175 isolation monitor.

Schaeffler Group

Troy, MI · Jun 2021 – Aug 2023

Chassis Mechatronics Engineer

- Modeled steering rack forces for steer-by-wire and wrote system requirements and specifications for a battery management system architecture.
- Supported prototype EV platforms, including the NAIAS 2023 demo vehicle, and presented them to customers, contributing to multi-million-dollar program awards.

> EDUCATION

B.S. Mechanical Engineering

Minor in Computer Science
Miami University · Oxford, OH · 2022

NCAA Division I Cross Country and Track & Field; team captain 2020–2022, conference title.

> SKILLS

Languages	Python, Rust, C++, C#, TypeScript, Bash, SQL, MATLAB
Networks	CAN / CAN FD, CANopen DS402, UDS, ISO-TP, J1939, LIN; DBC/SYM, MF4/TRC/MCAP
Embedded	Debian ARM64, systemd, U-Boot, UBI, RAUC OTA, WiFi/BT drivers, fleet provisioning
Test & cloud	DVPs, HIL, EOL test systems, root cause analysis; AWS, FastAPI, PostgreSQL, Docker/CI
Open source	bootCAN (Rust CAN/CAN FD analyzer), cantools contributor, DewDrop